

System Architecture — Agent Flow

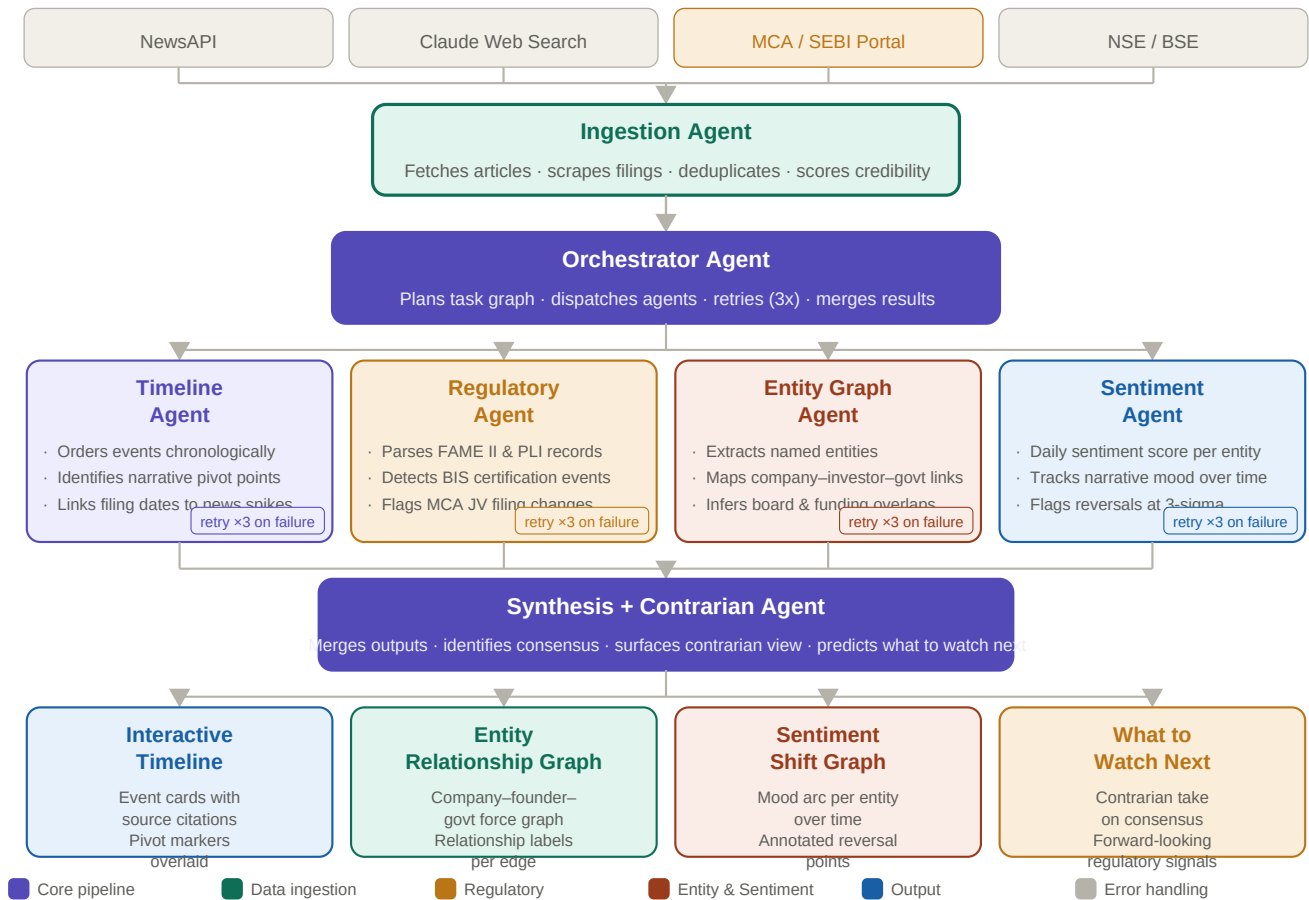


Figure 1. End-to-end agent pipeline. Data sources feed the Ingestion Agent, which passes structured data to the Orchestrator. Four specialist agents run in parallel. Their outputs merge into the Synthesis Agent, which delivers the four-panel dashboard.

Agent Roles & Communication

Agent	Role	Input	Output
Ingestion	Fetches articles via NewsAPI and Claude web search. Scrapes regulatory filings from MCA and SEBI. Deduplicates by content hash. Scores source credibility.	Topic string	Structured article and filing list
Timeline	Orders events chronologically. Identifies narrative pivot points — tone reversals, policy changes. Links regulatory filing dates to corresponding news spikes.	Article list	Ordered event list with pivot flags
Regulatory	Parses FAME II disbursements, PLI scheme rounds, BIS safety certifications, and MCA JV filings. Estimates financial impact per event.	Filing list	Regulatory event list with dates and impact
Entity Graph	Extracts named entities: companies, founders, ministers, investors. Infers relationships from co-occurrence. Maps government-corporate links and board overlaps.	Article list	Node-edge graph JSON
Sentiment	Sentence-level sentiment scoring on every article. Aggregates daily score per entity. Detects reversal points using a 3-sigma threshold.	Article list	Time-series sentiment data with reversal annotations
Synthesis + Contrarian	Merges all agent outputs. Identifies dominant media consensus. Generates the contrarian counter-argument the consensus is missing. Produces forward-looking signals.	All agent outputs	Unified story brief JSON delivered to UI

Error Handling & Audit Trail

Failure scenario	Orchestrator behaviour	Audit log entry
Agent times out (>30s)	Retry up to 3x with exponential backoff. On third failure, skip agent and flag the gap clearly in the UI.	AGENT_TIMEOUT · agent_id · attempt_count
NewsAPI rate limit hit	Automatically switch to Claude web search tool. Continue pipeline without interruption.	SOURCE_FALLBACK · reason · fallback_used
Two or more specialist agents fail	Abort synthesis. Return partial results to UI with a clear 'incomplete data' indicator.	PIPELINE_ABORT · agents_failed · partial_data
Synthesis agent fails	Retry once. On second failure, surface raw agent outputs directly to the dashboard.	SYNTHESIS_FAIL · fallback_mode = raw_outputs

Tool Integrations

Tool / API	Agent	Purpose	Fallback
NewsAPI	Ingestion	Fetch articles by keyword and date range	Claude web search tool
Claude web search	Ingestion	Real-time signals, press releases, gazette notifications	Log missing sources in audit trail
MCA portal	Regulatory	JV filings, director appointments, subsidiary changes	Cached filing dataset
Claude Sonnet API	All agents	Summarisation, entity extraction, sentiment scoring, synthesis	Retry 3x · abort on persistent failure
FastAPI + Streamlit	Orchestrator / UI	REST API backend serving four-panel interactive dashboard	Core infrastructure — no fallback